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RNN-AGT: A Recurrent Neural Network Framework for Accelerated Gap-Time Modeling via Gehan-Type Rank Loss

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ABSTRACT

Recurrent event data arise when the same type of outcome occurs repeatedly for a given subject over time, for example repeated hospitalizations, successive equipment failures, or recurring insurance claims. Standard statistical models for such data often rely on restrictive assumptions about how patient characteristics influence event risk, and can perform poorly when those assumptions are violated or when relationships between covariates and outcomes are complex. We propose **RNN-AGT**, a deep learning framework that predicts the time between successive recurrent events in a flexible, data-driven way. The model uses a recurrent neural network architecture to learn how a subject's event history shapes future event timing, without imposing parametric assumptions on the error distribution or requiring covariate effects to be linear. To handle subjects whose follow-up ends before all events are observed, a common feature of clinical and observational data, training is based on a rank-based objective that appropriately accounts for incomplete observations while remaining computationally efficient. We evaluate the method through extensive simulation experiments covering a wide range of scenarios, including nonlinear covariate effects, complex within-subject dependence, varying degrees of incomplete follow-up, and high-dimensional predictor spaces. The method is also applied to two clinical datasets involving repeated infections and hospital readmissions, and compared against established recurrent-event models. RNN-AGT demonstrates strong predictive accuracy and discrimination across all settings and outperforms classical competitors on the larger dataset, illustrating its potential as a practical tool for modeling recurrent events in biomedical and other applied contexts.

1 | Clinical Motivation

Many of the most pressing problems in clinical medicine involve not a single adverse event but a recurring one: the same patient is hospitalized repeatedly, contracts successive infections, or returns to the emergency department again and again. In these settings, the question clinicians and health systems need answered is not merely whether the next event will happen, but *when*, and how that timing changes with treatment, patient history, and the accumulating burden of prior episodes. Accurate answers directly shape clinical decisions: when to intensify a preventive therapy, when to schedule the next follow-up visit, and where to direct limited resources before the next crisis rather than after it.

Chronic obstructive pulmonary disease illustrates the stakes. Acute exacerbations occur on average 1.5 to 2.5 times per year in patients with moderate-to-severe disease, and each episode substantially increases the risk of the next [1, 2]. The interval between successive exacerbations tends to shorten as the disease progresses, and a patient's history of prior episodes, their treatment regimen, and their lung-function trajectory all interact to determine when the next hospitalization will occur. Knowing which patients are on the fastest trajectory toward their next admission would allow clinicians to act preventively rather than reactively, yet current risk tools are too coarse to capture this individual-level dynamics.

Recurrent infections in immunocompromised patients present a similar challenge. In a placebo-controlled trial of prophylactic gamma interferon involving 128 patients with Chronic Granulomatous Disease, participants experienced between zero and seven serious infections over a median follow-up of 293 days [3]. The central clinical question is whether the prophylaxis meaningfully

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delays recurrence and, if so, whether that benefit is uniform across patients differing in age, sex, and infection history. A treatment that reduces infection frequency in the average patient may still leave a high-risk subgroup largely unprotected, and identifying that subgroup requires a model sensitive to how a patient's prior course shapes their future risk.

Hospital readmissions after colorectal cancer surgery carry analogous consequences. Patients may return multiple times following resection for complications ranging from wound infection to bowel obstruction, with each readmission worsening quality of life and straining hospital capacity [4]. The timing of the next readmission depends on how far the disease has progressed, whether chemotherapy was administered, and the pattern of earlier readmissions. Patients with advanced disease tend to return sooner and more often, but even within a given disease stage, individual trajectories vary substantially. Predicting those trajectories at the individual level would help discharge teams plan follow-up care and flag patients likely to need early readmission support.

Across all three settings, the clinical need is the same: a tool that takes everything known about a patient (their characteristics, their treatment, and their history of prior events) and translates it into a reliable forecast of when the next event is likely to occur. Building and validating such a tool is the purpose of this paper.

2 | Introduction

Recurrent events constitute a ubiquitous structural feature of longitudinal data in medicine, engineering, economics, and the actuarial sciences, with particularly high stakes in clinical settings where the timing between successive events directly governs treatment decisions and resource allocation (Section 1). The scientific objective common to all such settings is to characterize the temporal dynamics of event occurrence: predicting future recurrences, evaluating interventions, and enabling efficient allocation of clinical resources. These tasks demand models capable of representing complex, history-dependent patterns in event timing.

The classical inferential framework for recurrent events is built upon hazard-based regression. The landmark Cox proportional hazards model [5] and its principled recurrent-event extensions (the Andersen–Gill counting-process formulation [6], the Prentice–Williams–Peterson gap-time and total-time models [7], and the Wei–Lin–Weissfeld marginal approach [8]) offer interpretable, well-characterized inferential procedures grounded in counting-process martingale theory. However, each rests on a multiplicative hazard structure with a log-linear predictor, an assumption whose violation is ubiquitous in modern applications characterized by high-dimensional feature sets, interaction effects, and nonstationary event dynamics [9]. When proportionality fails or the true predictor is genuinely nonlinear, these models yield biased coefficient estimates and degraded out-of-sample predictive performance, a limitation of increasing practical consequence as investigators turn to richer longitudinal datasets.

Accelerated failure time (AFT) models offer a complementary and, in many respects, more interpretable regression paradigm: rather than specifying a multiplicative structure on the instantaneous failure rate, they posit a direct multiplicative relationship between covariate values and the event-time scale itself [10]. This formulation permits survival-time prediction under arbitrary hazard shapes, and its regression coefficients admit a natural time-acceleration interpretation that is often more clinically meaningful than a log-hazard ratio. Classical semiparametric AFT estimation proceeds via rank-based estimating equations [11, 12, 13], which circumvent any distributional assumption on the error term at the cost of non-smoothness in the objective. For single-event survival data, the rank-based AFT methodology is now well-developed; see Wei, Ying, and Lin [14], Jin, Lin, Wei, and Ying [15], and Strawderman [16] for large-sample theory, and Jin, Lin, and Ying [17] for resampling-based variance estimation. Extensions to recurrent gap times were established by Lyu, Luo, Xu, and Huang [18], who proved consistency and asymptotic normality for a class of induced-smoothed rank estimators within the weighted risk-set (WRS) framework of Luo and Huang [19], wherein each subject's contribution is normalized by their event count so that subjects with many observed events do not disproportionately govern inference.

Despite these advances, both the hazard-based and the rank-based AFT paradigms for recurrent events share a fundamental expressiveness limitation: the predictor is constrained to be *linear* in the covariates, precluding the discovery of interaction effects, smooth nonlinear dose-response curves, and temporally evolving history dependence of the type that characterizes many real clinical sequences. The rapid growth of deep learning has catalyzed a parallel literature on neural-network-based survival models designed to overcome precisely this constraint. For single-event data, DeepSurv [20] embeds a multilayer perceptron within the Cox partial likelihood, while DeepAFT [21] incorporates neural predictors into Buckley–James and IPCW-reweighted AFT losses. The rank-based deep AFT approach of DeepR-AFT [22] leverages Gehan-type pairwise weighting [23] and has demonstrated competitive performance relative to random survival forests and parametric AFT models. Sequential architectures have been explored for single-event survival through RNN-SURV [24], Deep Recurrent Survival Analysis (DRSA) [25], and LSTM-Cox [9], each coupling a recurrent encoder with a proportional-hazards readout head. Transformer-based architectures have also emerged, exploiting multi-head self-attention to model long-range temporal dependencies without the inductive bias of recurrence [26].

Nevertheless, neural survival models specifically designed for *recurrent gap-time* prediction remain conspicuously absent from the literature. The naïve strategy of treating successive gap times as independent observations ignores within-subject temporal dependence and leads to invalid inference [19]. Subject-level frailty models partially redress this concern but impose distributional assumptions on the random effect and cannot capture complex sequential dynamics in the gap-time process. To date, no unified model simultaneously addresses (i) nonlinear and history-dependent covariate effects, (ii) weighted risk-set normalization to correct for unequal event counts, and (iii) incomplete follow-up arising from a finite monitoring horizon, all within a computationally tractable deep learning framework.

The present paper fills this gap by introducing **RNN-AGT** (**R**ecurrent **N**eural **N**etwork for **A**ccelerated **G**ap-**T**imes), a framework that embeds the AFT paradigm within a multi-layer GRU architecture purpose-built for recurrent gap-time prediction. The principal contributions of this work are fourfold:

- **A nonlinear AFT framework for recurrent gap times.** We extend the semiparametric AFT gap-time model of Lyu, Luo, Xu, and Huang [18] by replacing the linear predictor with a history-dependent nonlinear function learned by a stacked GRU network, enabling flexible representation of temporal dependence, covariate interactions, and smooth response surfaces that are inaccessible to classical linear AFT estimation.
- **A scalable rank-based training objective.** We derive the Gehan-type WRS loss for the nonlinear setting and prove that a mini-batch importance-weighted subsampling scheme yields an unbiased estimator of the full pairwise loss (Theorem A.2), reducing per-epoch complexity while preserving the statistical properties of the full-data estimator.
- **Theoretical support for the evaluation metric.** We establish large-sample consistency of the IPCW concordance index in the recurrent gap-time setting (Theorem A.3), providing a rigorous foundation for measuring predictive discrimination under incomplete follow-up with non-informative monitoring-time distributions.
- **Empirical validation.** Through extensive simulation experiments spanning linear, interaction-based, and GAM-type nonlinear mean structures, varying degrees of incomplete follow-up, two within-subject dependence mechanisms, and covariate spaces up to $p = 1{,}000$, together with analysis of two clinical datasets, we demonstrate that RNN-AGT achieves strong predictive accuracy and matches or outperforms classical Cox-based recurrent-event competitors.

The remainder of the paper is organized as follows. Section 3 develops the RNN-AGT framework in full: the recurrent gap-time data structure (Section 3.1), the AFT specification and its nonlinear GRU extension (Section 3.2), the GRU architecture (Section 3.3), the Gehan-type WRS rank loss and its nonlinear generalization (Section 3.4), and the mini-batch subsampling scheme (Section 3.5). Section 4 reports finite-sample performance across a comprehensive range of simulation settings. Section 5 presents the real-data analysis and benchmark comparisons. Section 6 concludes with a discussion of limitations and directions for future research. All technical proofs appear in the Appendix.

3 | Methodology

Let $i = 1, \dots, n$ index subjects and let $j \geq 1$ index successive recurrent events for subject i. Denote by S_{ij} the calendar time of the jth event, with $S_{i0} = 0$ by convention, and define the jth *gap time* as $T_{ij} = S_{ij} - S_{i,j-1}$. The overarching objective is to model and predict the sequence $\{T_{ij}\}_{j\geq 1}$ in a setting where (a) covariate effects on the log-gap-time scale may be nonlinear and history-dependent, (b) successive events for the same subject exhibit temporal dependence of an unspecified form, and (c) observation of gap times is subject to incomplete follow-up induced by a finite monitoring horizon.

RNN-AGT achieves this by coupling an accelerated failure time specification on the log-gap-time scale with a multi-layer GRU architecture that summarizes event history through a sequence of evolving latent state vectors, and by training the resulting model via a Gehan-type rank-based loss that accommodates incomplete follow-up without imposing any parametric distributional assumptions.

3.1 | Recurrent Gap-Time Data Structure

In practice, subject follow-up terminates at a monitoring time C_i. Let

$$K_i = \max\left\{k \geq 0 : \sum_{j=1}^{k} T_{ij} \leq C_i\right\}$$

denote the number of *fully observed* gap times for subject i. The observable gap time at event index j is

$$G_{ij} = \min\left(T_{ij},\ C_i - \sum_{l=1}^{j-1} T_{il}\right),$$

which equals T_{ij} when the jth gap time is complete and collapses to the remaining monitoring window otherwise. The associated event indicator is

$$\delta_{ij} = \mathbf{1}\left(\sum_{l=1}^{j} T_{il} \leq C_i\right),$$

so $\delta_{ij} = 1$ signals a fully observed gap time and $\delta_{ij} = 0$ signals an incompletely observed gap time (i.e., the subject's monitoring window closed before the jth event occurred). For notational convenience, define $Y_{ij} = \log G_{ij}$ and $K_i^* = \max(K_i, 1)$, where the latter retains at least one record per subject even when no complete event occurs before C_i.
The observed data for subject i are compactly written as

$$\mathcal{O}_i = \{(G_{ij}, \delta_{ij}) : j = 1, \dots, K_i^+;\ \mathbf{Z}_i, C_i\}, \qquad K_i^+ := K_i + 1,$$

where $\mathbf{Z}_i \in \mathbb{R}^p$ is the vector of baseline covariates and the final index K_i^+ captures the (possibly incomplete) gap time straddling the monitoring horizon.

Throughout, we maintain the standard assumption that C_i is independent of $\{T_{ij}, j \geq 1\}$ conditional on $\mathbf{Z}_i$ (non-informative end-of-monitoring). This condition, which is conventional in recurrent-event analysis [18, 19], guarantees that the marginal survival function of the monitoring time, $G_C(t) := \mathbb{P}(C_i \geq t)$, can be estimated consistently from the observed data via the Kaplan–Meier estimator applied to $\{C_i\}$, a quantity that plays a central role in both the IPCW evaluation metrics of Section 4.2 and the theoretical guarantees of Section A.

3.2 | AFT Specification for Recurrent Gap Times

3.2.1 | Linear Baseline

The semiparametric AFT model for recurrent gap times studied by Lyu, Luo, Xu, and Huang [18] postulates the log-linear relation

$$Y_{ij} = \theta_0' \mathbf{Z}_i + \varepsilon_{ij}, \qquad j \geq 1, \tag{3.1}$$

where $\theta_0 \in \mathbb{R}^p$ is the true coefficient vector and the error sequence $\{\varepsilon_{ij}\}_{j\geq 1}$ shares a common unspecified marginal distribution across event indices. Crucially, no parametric family is imposed on this marginal distribution, and no restriction is placed on the within-subject dependence structure of $(\varepsilon_{i1}, \varepsilon_{i2}, \ldots)$. This semiparametric agnosticism accommodates autoregressive, frailty-induced, and mixed dependence structures on equal footing, and stands in sharp contrast to shared-frailty models, which require conditional independence of gap times given the latent random effect.

3.2.2 | Nonlinear Extension via Stacked GRUs

The central methodological contribution of this work is the replacement of the linear predictor $\theta_0'\mathbf{Z}_i$ in (3.1) by a flexible, history-dependent function $\widehat{\mu}_{ij}(\mathbf{w})$ parameterized by a stacked GRU network with trainable weight collection $\mathbf{w}$. The resulting observation equation on the log-gap-time scale is

$$Y_{ij} = \widehat{\mu}_{ij}(\mathbf{w}) + \varepsilon_{ij}, \qquad j \geq 1,$$

where the error term ε_{ij} retains the semiparametric structure of the linear baseline: its marginal distribution is unspecified, and no assumptions govern the within-subject dependence beyond existence of a common marginal.
The network processes event sequences through L stacked recurrent layers. Let $\mathbf{h}_{ij}^{(\ell)} \in \mathbb{R}^{d_\ell}$ denote the hidden state at layer ℓ and event index j for subject i. The layer-wise recurrence reads

$$\mathbf{h}_{ij}^{(\ell)} = \Psi_{\mathbf{w}}^{(\ell)}\big(\mathbf{h}_{i,j-1}^{(\ell)},\ \mathbf{x}_{ij}^{(\ell)}\big), \qquad \ell = 1, \ldots, L,$$

where $\mathbf{x}_{ij}^{(1)} = \mathbf{x}_{ij}$ is the observed input at layer 1 and $\mathbf{x}_{ij}^{(\ell)} = \mathbf{h}_{ij}^{(\ell-1)}$ for $\ell \geq 2$. A linear output head applied to the terminal layer then yields the predicted log-gap-time mean:

$$\widehat{\mu}_{ij}(\mathbf{w}) = \mathbf{w}_{\text{out}}'\mathbf{h}_{ij}^{(L)} + b_{\text{out}},$$

constituting an estimate of $\mathbb{E}[Y_{ij} \mid \mathcal{F}_{i,j-1}]$, the conditional mean of the log-gap time given the filtration $\mathcal{F}_{i,j-1}$ generated by the observed history through event $j-1$.

3.2.3 | Covariate Initialization

Baseline covariates are encoded once into the initial hidden state of the first recurrent layer via the deterministic nonlinear projection

$$\mathbf{h}_{i0}^{(1)} = \tanh\big(W_{\text{init}}\mathbf{Z}_i + \mathbf{b}_{\text{init}}\big), \quad \mathbf{h}_{i0}^{(\ell)} = \mathbf{0}, \quad \ell = 2, \ldots, L.$$

This *covariate encoding strategy* cleanly separates static baseline information from the dynamic event history summarized by the recurrent layers. Encoding the covariates into the initial state rather than concatenating them to the input at each event index reduces the risk of information dilution in long sequences and improves computational efficiency by avoiding redundant feature propagation. For the sequential input at event j, we take the previous log-observable gap time $\mathbf{x}_{ij} = Y_{i,j-1}$; time-varying covariates $\mathbf{Z}_{ij}$ may be incorporated in a straightforward extension by appending them to the sequential input as $\mathbf{x}_{ij} = [Y_{i,j-1}', \mathbf{Z}_{ij}']'$.

3.3 | GRU Architecture

A fundamental challenge in training deep recurrent networks via backpropagation through time (BPTT) is the multiplicative accumulation of Jacobian matrices across the full sequence, a product that decays exponentially (vanishing gradient) or grows unboundedly (exploding gradient) as the sequence length increases [27, 28]. These pathologies impede the learning of long-range temporal dependencies, a critical concern in recurrent-event settings where the timing of early events may strongly condition subsequent gap-time distributions through complex patient-specific mechanisms.

Gated Recurrent Units [29] resolve these gradient flow issues through the introduction of two multiplicative gating mechanisms: the *update gate* $\mathbf{z}_{ij}^{(\ell)}$, which interpolates continuously between retaining the previous hidden state and adopting a newly computed candidate, and the *reset gate* $\mathbf{r}_{ij}^{(\ell)}$, which selectively suppresses stale historical information when forming the candidate state. At layer ℓ, the complete GRU update reads:

$$\begin{aligned}
\mathbf{z}_{ij}^{(\ell)} &= \sigma\big(W_z^{(\ell)}\mathbf{x}_{ij}^{(\ell)} + U_z^{(\ell)}\mathbf{h}_{i,j-1}^{(\ell)} + \mathbf{b}_z^{(\ell)}\big),\\
\mathbf{r}_{ij}^{(\ell)} &= \sigma\big(W_r^{(\ell)}\mathbf{x}_{ij}^{(\ell)} + U_r^{(\ell)}\mathbf{h}_{i,j-1}^{(\ell)} + \mathbf{b}_r^{(\ell)}\big),\\
\tilde{\mathbf{h}}_{ij}^{(\ell)} &= \tanh\big(W_h^{(\ell)}\mathbf{x}_{ij}^{(\ell)} + U_h^{(\ell)}\big(\mathbf{r}_{ij}^{(\ell)} \odot \mathbf{h}_{i,j-1}^{(\ell)}\big) + \mathbf{b}_h^{(\ell)}\big),\\
\mathbf{h}_{ij}^{(\ell)} &= \big(1 - \mathbf{z}_{ij}^{(\ell)}\big) \odot \mathbf{h}_{i,j-1}^{(\ell)} + \mathbf{z}_{ij}^{(\ell)} \odot \tilde{\mathbf{h}}_{ij}^{(\ell)},
\end{aligned} \tag{3.2a}$$

where $\sigma(\cdot)$ denotes the logistic sigmoid and $\odot$ the Hadamard (element-wise) product. The architecture at layer ℓ involves parameter matrices $W_z^{(\ell)}, W_r^{(\ell)}, W_h^{(\ell)} \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$ (input-to-hidden) and $U_z^{(\ell)}, U_r^{(\ell)}, U_h^{(\ell)} \in \mathbb{R}^{d_\ell \times d_\ell}$ (hidden-to-hidden), all learned jointly with the covariate initialization matrices $(W_{\text{init}}, \mathbf{b}_{\text{init}})$ and the linear output head $(\mathbf{w}_{\text{out}}, b_{\text{out}})$.

Relative to the Long Short-Term Memory (LSTM) network, the GRU employs two gates rather than three and maintains a single hidden state rather than separate cell and hidden states, reducing the total parameter count by approximately one-third while achieving comparable expressiveness on most sequence modeling benchmarks [30]. Stacking L GRU layers provides a hierarchical temporal representation: shallower layers learn local event-to-event patterns, while deeper layers encode progressively longer-range sequential dependencies. Dropout regularization may be applied between layers to mitigate overfitting. The unrolled multi-layer architecture is illustrated in Figure 1.

3.4 | Rank-Based Loss for Recurrent Gap Times

We now develop the training objective that renders RNN-AGT applicable to the incomplete gap-time observation structure described above. The development proceeds in three steps: we first recall the WRS rank-estimating function for the linear AFT model, then identify the corresponding convex loss whose subgradient it is, and finally extend both objects to the nonlinear predictor produced by the GRU network.

3.4.1 | Weighted Risk-Set Rank Estimating Equations

For the linear AFT model (3.1), define the residual on the log-gap-time scale as

$$e_{ij}(\theta) = Y_{ij} - \theta'\mathbf{Z}_i. \tag{3.3}$$

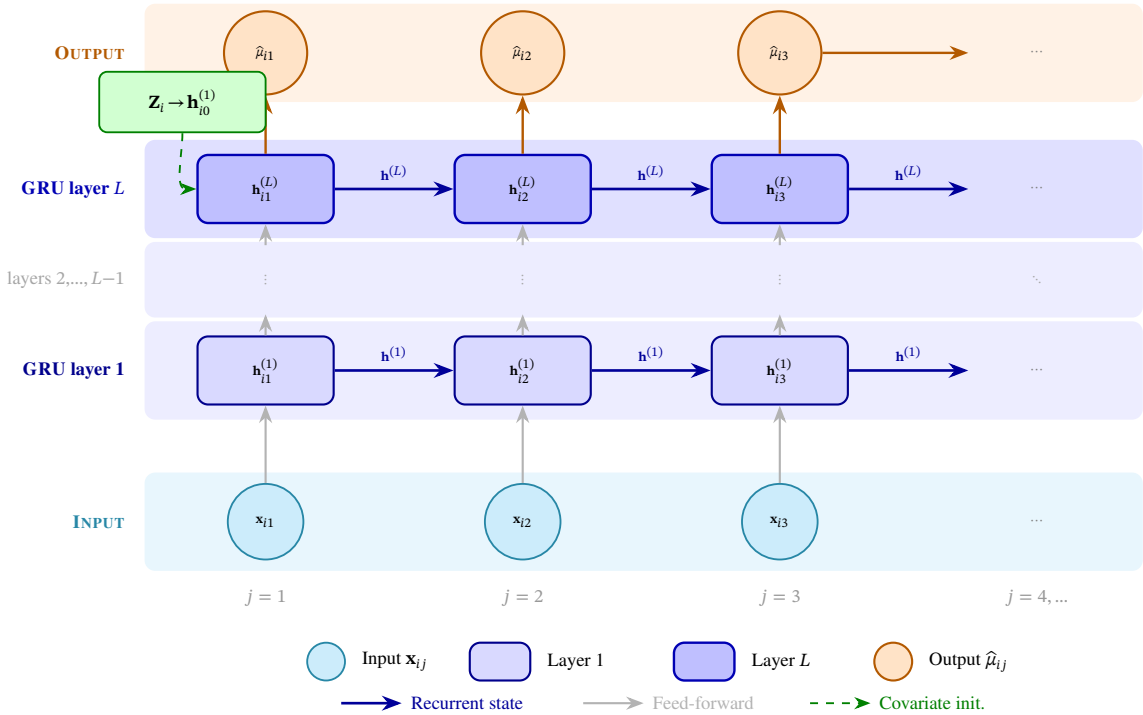


FIGURE 1 | Unrolled stacked GRU architecture for a fixed subject i. Each **teal circle** is a sequential input $\mathbf{x}_{ij} = Y_{i,j-1}$ (the previous log-gap time). **Blue rectangles** are GRU hidden states that propagate recurrently across event indices (horizontal arrows) and feed forward to the next layer (vertical arrows). **Darker blue** marks the terminal GRU layer L whose states feed directly into the **orange output heads** $\hat{\mu}_{ij}$. The **green dashed** callout indicates that baseline covariates $\mathbf{Z}_i$ are injected once via the initial hidden state $\mathbf{h}_{i0}^{(1)}$ rather than repeated at every event index.

Following Lyu, Luo, Xu, and Huang [18], introduce subject-level averaged counting and at-risk processes with WRS normalization:

$$N_i^*(\theta, t) = \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} \delta_{ij} \mathbf{1}\{e_{ij}(\theta) \le t\}, \quad R_i^*(\theta, t) = \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} \mathbf{1}\{e_{ij}(\theta) \ge t\}.$$

The $1/K_i^*$ weighting is the statistical heart of the WRS construction: it ensures that each subject contributes comparably to the estimating equation regardless of the length of their observed event history, thereby preventing subjects with many events from dominating inference. Aggregate risk-set summaries take the form

$$S_0^*(\theta, t) = \frac{1}{n} \sum_{i=1}^{n} R_i^*(\theta, t), \quad S_1^*(\theta, t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{Z}_i R_i^*(\theta, t).$$

The Gehan-type WRS estimating equation [23, 19] is then

$$U(\theta) = \sum_{i=1}^{n} \int S_0^*(\theta, t) \left\{ \mathbf{Z}_i - \frac{S_1^*(\theta, t)}{S_0^*(\theta, t)} \right\} dN_i^*(\theta, t) = \mathbf{0},$$

where the Gehan weight $W^* = S_0^*$ downweights comparisons in sparse risk sets, providing improved finite-sample stability relative to the unweighted log-rank statistic and connecting naturally to a convex loss function.

3.4.2 | Induced Convex Loss

As established by Fygenson and Ritov [12] for single-event data and extended to the WRS recurrent gap-time framework by Lyu, Luo, Xu, and Huang [18], the estimating function $U(\theta)$ is (up to sign) the subgradient of the convex, non-smooth objective

$$L_G(\theta) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \sum_{l=1}^{n} \sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} \left[e_{lk}(\theta) - e_{ij}(\theta) \right]^-, \tag{3.4}$$

where $[a]^- = \max(0, -a)$ denotes the hinge (negative part) function. The loss accumulates, over all ordered pairs of retained gap-time records, the extent to which the residual of a comparison record (l, k) falls *below* that of a fully observed anchor record (i, j); the indicator δ_{ij} restricts the anchor role to completely observed gap times, appropriately accounting for the incomplete observation structure. Minimizing $L_G(\theta)$ is equivalent to solving the estimating equation $U(\theta) = \mathbf{0}$ in the sense that their solution sets coincide under standard regularity conditions.

Remark 1. Connection to single-event AFT. When $K_i^* \equiv 1$ for all i, the WRS normalization is trivially unity and $L_G(\theta)$ reduces to the standard Gehan-type rank loss employed in single-event AFT estimation [23, 22].

3.4.3 | Nonlinear Extension

Theorem A.1 in the Appendix formalizes the extension of L_G to the nonlinear predictor $\hat{\mu}_{ij}(\mathbf{w})$ produced by the GRU network. Replacing the linear residual (3.3) with the induced nonlinear residual $e_{ij}(\mathbf{w}) = Y_{ij} - \hat{\mu}_{ij}(\mathbf{w})$ in (3.4) yields the RNN-AGT training objective:

$$L_G^{\mathrm{NL}}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \sum_{l=1}^{n} \sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} \left[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w}) \right]^-.$$

Three features of L_G^{NL} merit emphasis. First, the loss is inherently *rank-based*: it depends only on the relative ordering of residuals, not their absolute magnitudes, conferring robustness to heavy-tailed error distributions and outliers that would distort squared-error objectives. Second, unlike $L_G(\theta)$, $L_G^{NL}(\mathbf{w})$ is *not convex* in $\mathbf{w}$ because the residual map $\mathbf{w} \longmapsto \widehat{\mu}_{ij}(\mathbf{w})$ is a nonlinear function of the network parameters; this necessitates stochastic gradient-based optimization rather than convex programming. Third, the non-smoothness of $[\cdot]^-$ introduces kinks in the loss landscape, so gradients are interpreted in the subgradient sense, with the convention that $\partial[a]^-/\partial a = 0$ at $a = 0$.

3.5 | Mini-Batch Subsampling for Scalable Training

Let $N^* = \sum_{i=1}^n K_i^*$ denote the total number of retained gap-time records and $n' = \sum_{i,j} \delta_{ij}$ the number of fully observed records. A direct evaluation of $L_G^{NL}(\mathbf{w})$ requires $O(N^{*2})$ pairwise comparisons per gradient step, a quadratic cost that rapidly becomes prohibitive when subjects experience many events or sample sizes are large. We address this bottleneck through a principled importance-weighted subsampling scheme.

3.5.1 | Subsampled Estimator

Let $\mathcal{I} = \{(i,j) : j = 1, \dots, K_i^*\}$ and $\mathcal{U} = \{(i,j) \in \mathcal{I} : \delta_{ij} = 1\}$ (cardinality n'). For each anchor $(i,j) \in \mathcal{U}$, draw s comparison indices $(l,k) \sim q_{ij}$ i.i.d. from a positive sampling distribution on $\mathcal{I}$, yielding a sampled pair set $\mathcal{P}_s$ of cardinality $m = n's$. The resulting importance-weighted estimator is

$$\widetilde{L}_{G,s}^{NL}(\mathbf{w}) = \frac{1}{n} \sum_{(i,j)\in\mathcal{U}} \frac{1}{s} \sum_{r=1}^{s} \frac{\left[e_{(l,k)_{ij,r}}(\mathbf{w}) - e_{ij}(\mathbf{w})\right]^-}{K_i^* K_{l_r}^* q_{ij}((l,k)_{ij,r})}, \tag{3.5}$$

where $K_{l_r}^*$ denotes K_l^* for the rth sampled index $(l,k)_{ij,r}$. Theorem A.2 (Appendix) establishes that $\mathbb{E}[\widetilde{L}_{G,s}^{NL}(\mathbf{w}) \mid \mathcal{O}] = L_G^{NL}(\mathbf{w})$ for every $\mathbf{w}$, so stochastic gradient descent on $\widetilde{L}_{G,s}^{NL}$ constitutes an asymptotically valid surrogate for optimizing the full pairwise loss. Under the uniform choice $q_{ij}(l,k) = 1/N^*$, the importance weight is constant and the estimator simplifies to an unweighted average of m sampled pairwise terms, reducing per-epoch complexity from $O(N^{*2})$ to $O(n's)$.

3.5.2 | Mini-Batch Optimization

In practice, stochastic parameter updates are computed over mini-batches $\mathcal{B} \subset \mathcal{P}_s$ of size b:

$$\widetilde{L}_{G,\mathcal{B}}^{NL}(\mathbf{w}) \propto \sum_{((i,j),(l,k))\in\mathcal{B}} \frac{\delta_{ij}}{K_i^* K_l^*} \omega_{ij}(l,k) \left[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w})\right]^-,$$

where $\omega_{ij}(l,k) = 1/q_{ij}(l,k)$ is the importance weight. Parameters are updated via

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha \nabla_{\mathbf{w}} \widetilde{L}_{G,\mathcal{B}}^{NL}(\mathbf{w}),$$

using an adaptive optimizer such as Adam [31] or RMSprop.

Remark 2. Computational complexity. The per-epoch cost of mini-batch optimization scales as $O(n's)$, compared with $O(N^{*2})$ for the full Gehan loss. For $K_i^* \sim$ Poisson(2), this represents a reduction by a factor of roughly N^*/s. With $n = 1{,}000$, $\bar{K}^* = 2$, and $s = 10$, the sub-sampled cost is approximately $n's = 10{,}000$ pair evaluations versus $N^{*2} \approx 4{,}000{,}000$ for the full loss, a 400-fold reduction that makes large-scale recurrent-event modeling computationally tractable without sacrificing statistical validity.

4 | Simulation Studies

4.1 | Data-Generating Mechanisms

To evaluate RNN-AGT over a comprehensive and clinically representative range of scenarios, we generated recurrent gap-time data according to the log-linear model

$$y_{ij} = f(\mathbf{z}_i) + \varepsilon_{ij},$$

where $\mathbf{z}_i = (z_{i1}, z_{i2}, z_{i3})'$ are baseline covariates generated sequentially to induce moderate covariate correlation: $z_{i1} \sim$ Bernoulli(0.5), $z_{i2} \mid z_{i1} \sim \mathcal{N}(z_{i1}/3, 1)$, and $z_{i3} \mid z_{i2} \sim \mathcal{N}(z_{i2}/3, 1)$. This sequential construction yields Pearson correlations of approximately 0.30 between z_{i1} and z_{i2}, and 0.09 between z_{i1} and z_{i3}, mimicking the mild-to-moderate collinearity commonly encountered in clinical covariate sets while preserving a well-conditioned design.

Mean functions. We considered three mean functions of increasing structural complexity:

- **Nonlinear with interactions:** $f_1(\mathbf{z}) = 3z_1 + 7z_1z_2 - 5z_3$, featuring a binary-continuous interaction that cannot be captured by any additive linear predictor. This setting directly targets the interaction-discovery capability of GRU networks and constitutes the primary stress test of the nonlinear AFT extension;
- **GAM-type nonlinear:** $f_2(\mathbf{z}) = z_1 + z_2^3 + \exp(0.9z_3)$, incorporating smooth polynomial and exponential components that challenge the representational capacity of linear approximation. The polynomial component z_2^3 tests learning of non-monotone curvature, while $\exp(0.9z_3)$ tests rapid growth in the tail of a continuous predictor, both patterns common in dose-response and biomarker-risk modelling;
- **Linear:** $f_3(\mathbf{z}) = z_1 - 6z_2 + 4z_3$, providing the most favorable benchmark against which the nonlinear AFT generalizes. Inclusion of this setting verifies that RNN-AGT does not sacrifice performance relative to classical linear estimators when the linear model is correctly specified.

Error distributions. Error terms ε_{ij} were drawn from three standardized distributions (zero mean, unit variance): Normal, Gumbel, and Logistic. These three families were selected for distinct and complementary reasons. The *Normal* distribution serves as the symmetric light-tailed baseline and corresponds to the Weibull error structure under which parametric AFT models are most efficient, making it the most favorable setting for squared-error objectives. The *Logistic* distribution is also symmetric but has heavier tails (excess kurtosis = 1.2) and is the canonical error distribution for the log-logistic survival model; its heavier tails create outlier-prone residuals that test the robustness of the rank-based loss. The *Gumbel* distribution introduces right-skewness (skewness $\approx$ 1.14, excess kurtosis $\approx$ 2.4) and corresponds to the extreme-value error structure

of the log-Weibull (Gumbel minimum) model; it is the only asymmetric family in the design and probes robustness to distributional shape departures from symmetry. Together, the three families span symmetric light-tailed, symmetric heavy-tailed, and asymmetric heavy-tailed regimes. Because RNN-AGT's Gehan-type rank loss depends only on the *ordering* of residuals and not on their distributional family, good performance across all three distributions would confirm the semiparametric robustness claimed in Section 3.2.

Within-subject dependence. To reflect realistic within-subject correlation structure, two dependence mechanisms were employed:

- **Random-effects (frailty) model:** $\varepsilon_{ij} = b_i + \eta_{ij}$ with $b_i \sim \mathcal{N}(0, \sigma_b^2)$, $\sigma_b = 0.5$, and η_{ij} i.i.d. from the specified marginal error distribution. The frailty variance $\sigma_b^2 = 0.25$ yields an intraclass correlation of $\text{ICC} = \sigma_b^2/(\sigma_b^2 + 1) = 0.20$ under Gaussian innovations, representing moderate exchangeable within-subject dependence of the type induced by unmeasured patient-level random effects (e.g., genetic susceptibility, treatment adherence). This structure is the most common parametric assumption in shared-frailty recurrent-event models, providing a natural competitor for the GRU's history-dependent encoding;
- **Autoregressive AR(1) model:** $\eta_{ij} = \rho\,\eta_{i,j-1} + \sqrt{1-\rho^2}\,\xi_{ij}$ with $\rho = 0.5$ and ξ_{ij} i.i.d. from the specified marginal distribution. The coefficient $\rho = 0.5$ was chosen to produce moderate serial correlation (Pearson correlation between consecutive gap-time errors $= 0.5$, decaying geometrically as $0.5^{|j-k|}$ for lags $|j-k| \geq 1$) without approaching the near-unit-root regime that would make the process difficult to distinguish from nonstationarity at the sample sizes considered. Unlike the exchangeable frailty structure, AR(1) dependence is *sequential*: earlier gap times directly inform later ones through the lagged error, which is precisely the type of history dependence that the GRU recurrent architecture is designed to exploit. Comparing frailty and AR(1) results therefore directly quantifies the discriminative benefit of sequential modelling.

Incomplete follow-up. Monitoring times were generated as $C_i = \tau U_i$ with $U_i \sim \text{Uniform}(0,1)$, with the scalar $\tau > 0$ calibrated separately for each mean function to yield marginal incomplete follow-up proportions of approximately 25%, 50%, and 65%. The three levels were chosen to span practically relevant regimes: 25% incomplete follow-up is typical of well-followed cohort studies with short monitoring windows relative to inter-event times, 50% reflects moderate administrative truncation or loss to follow-up, and 65% represents a challenging high-incompleteness regime that stresses the IPCW reweighting and tests whether the rank-based objective can preserve discriminative ability when the majority of gap times are cut short by end of monitoring. Uniform monitoring times were adopted because they satisfy the non-informative end-of-monitoring assumption maintained throughout and yield a tractable closed-form for calibrating τ.

Event count and sample sizes. The number of gap times per subject was drawn from Poisson($\lambda = 2$), producing a mean of two events per subject before end of monitoring. This choice reflects the low-to-moderate event rates typical of clinical recurrent-event studies (e.g., hospitalizations in chronic disease cohorts) and ensures that the WRS normalisation $1/K_i^*$ is non-trivial, since subjects vary meaningfully in their observed event counts. Training sample sizes were $n_{\text{train}} \in \{1{,}000,\ 5{,}000\}$, representing a modest community-based study and a large registry, respectively. A fixed held-out test set of $n_{\text{test}} = 2{,}000$ was used across all settings. All results are averaged over $R = 500$ independent simulation replicates to ensure stable Monte Carlo estimates of the performance metrics.

Network architecture and training hyperparameters. The RNN-AGT model was implemented with $L = 2$ stacked GRU layers, each with hidden dimension $d = 64$, providing sufficient capacity to learn the nonlinear mean structures without excessive parameterisation at the sample sizes considered. Network weights were initialised using the Glorot uniform scheme [32], which keeps the variance of activations and gradients approximately constant across layers and is the standard initialisation for gated recurrent architectures. No dropout regularisation was applied in the main simulation experiments so that performance differences are attributable solely to the model and objective rather than to implicit regularisation; its effect is examined separately in the high-dimensional experiments of Section 4.3.4. Sub-sampling parameters were fixed at $s = 10$ pairs per fully observed anchor record and mini-batch size $b = 64$; the sensitivity of results to these choices is examined in Section 4.3. The Adam optimizer [31] was used with learning rate $\alpha = 10^{-4}$ and default momentum parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$. No learning rate schedule or early stopping was applied in the main tables; performance is instead reported at multiple epoch checkpoints (epochs 5, 10, 15 for $n_{\text{train}} = 1{,}000$ and epochs 2, 3, 5 for $n_{\text{train}} = 5{,}000$) to expose the full training trajectory and allow practitioners to identify optimal stopping points for their own settings.

4.2 | Evaluation Metrics

Performance was assessed via two complementary metrics capturing distinct aspects of predictive quality.

AMSE. The *adjusted mean squared error*, corrected for incomplete follow-up bias via inverse-probability-of-censoring weighting, is defined as

$$\text{AMSE} = \frac{1}{\sum_{i,j} K_i^*} \sum_{i=1}^{n_{\text{test}}} \sum_{j=1}^{K_i^*} \frac{\delta_{ij}}{\widehat{KM}(S_{ij})} (Y_{ij} - \widehat{\mu}_{ij})^2,$$

where $\widehat{KM}(t)$ is the Kaplan–Meier estimator of the monitoring-time survival function $G_C(t) = \mathbb{P}(C \geq t)$. The IPCW weight $1/\widehat{KM}(S_{ij})$ upweights the squared errors of fully observed events in proportion to their probability of not having been cut short by end of monitoring, yielding an approximately unbiased estimate of the expected squared prediction error under the complete-data distribution. When $G_C \equiv 1$ (complete follow-up for all subjects), AMSE reduces identically to the standard mean squared error.

IPCW Concordance Index. The *inverse-probability-of-censoring weighted concordance index* is

$$\widehat{C}_{\text{IPCW}} = \frac{\sum_{i \neq k} \sum_{j=1}^{K_i^*} \sum_{l=1}^{K_k^*} \frac{\delta_{ij}}{\widehat{KM}(S_{ij})} \mathbf{1}\{G_{ij} < G_{kl}\}\, \mathbf{1}\{\widehat{\mu}_{ij} < \widehat{\mu}_{kl}\}}{\sum_{i \neq k} \sum_{j=1}^{K_i^*} \sum_{l=1}^{K_k^*} \frac{\delta_{ij}}{\widehat{KM}(S_{ij})} \mathbf{1}\{G_{ij} < G_{kl}\}}. \tag{4.1}$$

This statistic measures the proportion of cross-subject comparable pairs for which the model correctly orders the gap times. Theorem A.3 (Appendix) establishes that $\widehat{C}_{\text{IPCW}}$ converges in probability to the true concordance probability $\mathbb{P}(\widehat{\mu}_{ij} < \widehat{\mu}_{kl} \mid T_{ij} < T_{kl})$ under non-informative end-of-monitoring, providing rigorous theoretical justification for its use as an evaluation criterion. Values of 0.5 and 1.0 correspond to random ordering (no discriminative ability) and perfect discrimination, respectively.

4.3 | Results

4.3.1 | Mean Function Estimation

Tables 1–3 report IPCW C-index and AMSE (formatted as C-index / AMSE pairs) under all combinations of mean function, error distribution, incomplete follow-up level, dependence structure, training sample size, and training duration.

4.3.1.1 | Nonlinear with interactions (Table 1). RNN-AGT consistently achieves high IPCW C-indices in the range 0.85–0.94, demonstrating strong discriminative ability even at 65% incomplete follow-up and under both frailty and AR(1) within-subject dependence. As expected, AMSE rises with the proportion of incompletely observed gap times, reflecting the reduced effective sample size available for IPCW reweighting; the concordance index, however, remains strikingly stable across incomplete follow-up levels, confirming that the rank-based training objective successfully preserves relative risk ordering even when absolute point prediction is degraded by end-of-monitoring truncation. For $n_{\text{train}} = 1{,}000$, extending training beyond 10–15 epochs yields diminishing or negative returns under high proportions of incomplete follow-up, indicating mild overfitting; for $n_{\text{train}} = 5{,}000$, near-optimal performance is typically achieved by epochs 3 to 5. Gumbel errors yield the most favorable AMSE values, while Logistic errors prove the most challenging. Notably, AR(1) dependence consistently outperforms the frailty structure in C-index, suggesting that the GRU recurrent architecture is particularly effective at exploiting serially structured within-subject patterns relative to exchangeable frailty-induced correlation.

4.3.1.2 | GAM-type nonlinear (Table 2). The smooth nonlinear mean structure presents a harder estimation problem, as the global polynomial and exponential components resist local linear approximation. C-indices range from approximately 0.63 to 0.87 for $n_{\text{train}} = 1{,}000$ and improve substantially at $n_{\text{train}} = 5{,}000$, underscoring the importance of adequate sample size for learning smooth nonlinear response surfaces. The sensitivity to training duration is more pronounced in this setting: under 50–65% incomplete follow-up, later epochs frequently decrease the C-index despite modest AMSE reductions, a pattern consistent with overfitting of the squared-error component while rank ordering deteriorates. These results strongly motivate the use of early stopping based on held-out concordance monitoring in this regime.

4.3.1.3 | Linear mean (Table 3). The linear setting constitutes the most favorable environment, with C-indices typically exceeding 0.90 and frequently surpassing 0.95 under Normal and Gumbel errors, even at 65% incomplete follow-up. The concordance index is stable across epochs at the discrimination level, while AMSE continues to inflate with training duration, reflecting the tendency of the rank-based objective to preserve relative ordering at the expense of calibrated magnitude prediction as training proceeds. This decoupling reinforces the practical importance of monitoring both metrics jointly during model development.

Across all settings, RNN-AGT maintains strong discriminative performance, adapts efficiently to varying sample sizes, and demonstrates robustness to diverse error distributions and within-subject dependence structures.

4.3.2 | Training Dynamics

To characterize the optimization behavior of RNN-AGT, we tracked the mini-batch training loss over 30 epochs under all three mean functions and error distributions, fixing $n_{\text{train}} = 1{,}000$ and 25% incomplete follow-up (Figure 2).

Across all nine configurations (three mean functions × three error distributions), the loss exhibits a stable and nearly monotone decrease, with no evidence of divergence or oscillation. The linear mean function converges most rapidly, reaching near-minimum values by epochs 5–8; the GAM-type nonlinear structure displays moderately slower convergence; and the interaction-based nonlinear mean function exhibits the highest initial loss values and the most gradual early-stage descent, consistent with the greater structural complexity of the underlying learning problem. Differences across error distributions are relatively minor, though heavier-tailed distributions (Gumbel, Logistic) tend to produce higher initial loss values and marginally slower convergence, particularly in the nonlinear settings. These results collectively confirm that the GRU architecture, coupled with the mini-batch Gehan-type loss, trains stably and robustly across the full range of data-generating scenarios examined.

4.3.3 | Sensitivity to Sub-Sampling Parameters

Table 4 reports IPCW C-index and AMSE for all combinations of $s \in \{5, 10, 15\}$ pairs per anchor record and mini-batch sizes $b \in \{32, 64, 128\}$, under the interaction-based nonlinear mean function with Normal errors and 25% incomplete follow-up. These experiments isolate the sensitivity of predictive performance to the two principal algorithmic hyperparameters.

Increasing s consistently improves discriminative performance, with the C-index rising from approximately 0.83–0.88 at $s = 5$ to 0.90–0.94 at $s = 15$ across epochs and training sizes. This monotone improvement in discrimination comes at a proportional increase in per-epoch computational cost, scaling as $O(n's)$. Smaller mini-batch sizes ($b = 32$) achieve the highest C-indices, particularly at the smaller training scale, suggesting that the additional stochasticity of noisier gradient estimates acts as an implicit regularizer in this regime. Larger batches ($b = 128$) stabilize the gradient signal but converge more slowly in terms of concordance per epoch. The configuration $s = 10$, $b = 64$ provides a robust and computationally efficient compromise across all conditions and is adopted as the default in all subsequent experiments.

4.3.4 | Performance Under High-Dimensional Covariates

To assess robustness to the curse of dimensionality, we augmented the three signal covariates with progressively many noise variables,

TABLE 1 | Simulation results (IPCW C-index / AMSE) for the **nonlinear mean function with interactions**, under varying error distributions, incomplete follow-up proportion ($\%C$), and within-subject dependence structures. Each cell reports C-index / AMSE.

Error	Dependence	$\%C$	$n_{\text{train}} = 1{,}000$			$n_{\text{train}} = 5{,}000$		
			Epoch 5	**Epoch 10**	**Epoch 15**	**Epoch 2**	**Epoch 3**	**Epoch 5**
Normal	Frailty	25	0.849 / 7.58	0.881 / 6.52	0.909 / 5.81	0.879 / 6.42	0.903 / 6.35	0.931 / 7.13
		50	0.863 / 10.91	0.863 / 10.64	0.860 / 13.75	0.855 / 11.13	0.863 / 13.40	0.878 / 18.49
		65	0.891 / 23.02	0.895 / 19.80	0.878 / 18.85	0.899 / 17.69	0.880 / 19.56	0.871 / 23.84
	AR(1)	25	0.851 / 7.37	0.892 / 5.92	0.918 / 6.39	0.882 / 6.69	0.904 / 6.17	0.941 / 6.77
		50	0.874 / 13.39	0.877 / 11.86	0.886 / 13.73	0.864 / 11.96	0.877 / 13.09	0.903 / 18.00
		65	0.901 / 21.68	0.897 / 18.10	0.883 / 18.29	0.903 / 17.78	0.888 / 19.27	0.880 / 24.57
Gumbel	Frailty	25	0.852 / 8.56	0.887 / 5.84	0.905 / 4.77	0.884 / 6.22	0.905 / 5.14	0.930 / 4.14
		50	0.880 / 11.04	0.892 / 9.14	0.882 / 10.83	0.879 / 9.36	0.865 / 11.19	0.865 / 13.97
		65	0.904 / 20.57	0.902 / 14.83	0.907 / 15.27	0.900 / 14.58	0.900 / 15.25	0.859 / 19.72
	AR(1)	25	0.843 / 8.72	0.895 / 4.98	0.934 / 3.08	0.893 / 5.32	0.912 / 4.47	0.940 / 3.56
		50	0.884 / 9.80	0.899 / 8.07	0.902 / 8.24	0.884 / 8.20	0.887 / 10.07	0.906 / 12.42
		65	0.904 / 19.00	0.919 / 15.23	0.902 / 13.39	0.907 / 13.69	0.906 / 13.47	0.881 / 18.58
Logistic	Frailty	25	0.848 / 10.18	0.875 / 8.63	0.891 / 7.62	0.870 / 9.40	0.887 / 8.74	0.904 / 8.35
		50	0.878 / 15.36	0.874 / 13.42	0.845 / 15.72	0.863 / 13.58	0.838 / 16.08	0.817 / 19.43
		65	0.893 / 26.39	0.887 / 20.94	0.867 / 21.06	0.870 / 21.23	0.858 / 22.36	0.776 / 27.91
	AR(1)	25	0.933 / 9.24	0.884 / 8.04	0.898 / 7.50	0.882 / 8.60	0.898 / 8.39	0.920 / 8.14
		50	0.886 / 14.75	0.889 / 13.03	0.873 / 15.71	0.878 / 13.26	0.873 / 15.70	0.873 / 18.95
		65	0.898 / 25.38	0.912 / 21.07	0.900 / 20.36	0.893 / 20.69	0.886 / 21.50	0.841 / 27.30

TABLE 2 | Simulation results (IPCW C-index / AMSE) for the **GAM-type nonlinear mean function**. Layout as in Table 1.

Error	Dependence	$\%C$	$n_{\text{train}} = 1{,}000$			$n_{\text{train}} = 5{,}000$		
			Epoch 5	**Epoch 10**	**Epoch 15**	**Epoch 2**	**Epoch 3**	**Epoch 5**
Normal	Frailty	25	0.841 / 8.30	0.837 / 6.57	0.830 / 5.83	0.836 / 6.68	0.830 / 6.80	0.831 / 10.63
		50	0.827 / 11.58	0.770 / 10.28	0.722 / 9.15	0.794 / 9.99	0.732 / 9.61	0.685 / 12.64
		65	0.849 / 16.22	0.768 / 14.16	0.698 / 13.59	0.806 / 15.97	0.740 / 14.40	0.595 / 13.97
	AR(1)	25	0.856 / 7.89	0.860 / 6.08	0.861 / 5.41	0.860 / 6.30	0.859 / 6.41	0.859 / 9.29
		50	0.853 / 11.28	0.828 / 9.46	0.817 / 8.33	0.841 / 9.79	0.805 / 9.54	0.754 / 11.63
		65	0.874 / 16.73	0.825 / 14.72	0.785 / 13.48	0.855 / 16.25	0.810 / 15.59	0.690 / 14.04
Gumbel	Frailty	25	0.824 / 7.63	0.816 / 5.19	0.815 / 3.89	0.814 / 4.66	0.804 / 4.11	0.806 / 6.71
		50	0.789 / 8.47	0.735 / 6.67	0.697 / 5.40	0.751 / 6.30	0.689 / 5.32	0.629 / 6.48
		65	0.809 / 10.73	0.740 / 8.98	0.704 / 7.73	0.760 / 9.28	0.714 / 8.15	0.612 / 7.24
	AR(1)	25	0.844 / 7.16	0.848 / 4.89	0.850 / 3.55	0.850 / 4.45	0.847 / 3.74	0.851 / 5.57
		50	0.850 / 8.10	0.827 / 6.03	0.795 / 4.75	0.814 / 6.03	0.776 / 5.00	0.740 / 5.89
		65	0.872 / 9.84	0.810 / 8.60	0.792 / 7.17	0.826 / 8.71	0.797 / 7.40	0.666 / 7.43
Logistic	Frailty	25	0.787 / 7.62	0.774 / 6.34	0.752 / 5.92	0.746 / 6.14	0.726 / 6.14	0.704 / 9.66
		50	0.756 / 9.17	0.686 / 7.92	0.655 / 7.45	0.667 / 8.22	0.619 / 8.13	0.538 / 10.44
		65	0.781 / 11.74	0.689 / 10.53	0.632 / 10.09	0.690 / 11.77	0.607 / 11.10	0.526 / 11.83
	AR(1)	25	0.800 / 6.84	0.812 / 5.64	0.809 / 5.35	0.804 / 5.71	0.791 / 5.99	0.783 / 8.86
		50	0.762 / 7.40	0.756 / 7.32	0.724 / 6.93	0.764 / 7.53	0.699 / 7.27	0.617 / 9.20
		65	0.843 / 11.15	0.769 / 9.80	0.719 / 9.91	0.778 / 10.61	0.729 / 10.48	0.584 / 11.25

increasing p from 3 to 1,000. Noise covariates were generated by extending the sequential scheme and assigned zero regression coefficients, so they carry no information about gap-time outcomes. The training sample was $n_{\text{train}} = 1{,}000$ with $n_{\text{test}} = 2{,}000$, Gaussian errors, and 25% incomplete follow-up.

TABLE 3 | Simulation results (IPCW C-index / AMSE) for the **linear mean function**. Layout as in Table 1.

Error	Dependence	%C	$n_{\text{train}} = 1{,}000$			$n_{\text{train}} = 5{,}000$		
			Epoch 5	**Epoch 10**	**Epoch 15**	**Epoch 2**	**Epoch 3**	**Epoch 5**
Normal	Frailty	25	0.924 / 10.51	0.941 / 14.93	0.944 / 18.30	0.942 / 15.64	0.944 / 19.42	0.943 / 27.87
		50	0.941 / 22.94	0.924 / 25.56	0.897 / 32.49	0.940 / 25.78	0.908 / 34.07	0.902 / 46.16
		65	0.940 / 40.89	0.957 / 34.68	0.934 / 38.37	0.961 / 34.23	0.942 / 39.57	0.877 / 53.82
	AR(1)	25	0.924 / 10.79	0.948 / 14.75	0.950 / 18.20	0.949 / 15.18	0.951 / 19.77	0.951 / 27.76
		50	0.941 / 22.91	0.935 / 26.09	0.924 / 31.56	0.945 / 25.09	0.929 / 32.62	0.927 / 43.53
		65	0.941 / 40.26	0.961 / 33.57	0.942 / 38.19	0.962 / 34.07	0.944 / 39.41	0.902 / 53.12
Gumbel	Frailty	25	0.917 / 8.80	0.926 / 11.85	0.928 / 16.43	0.933 / 11.46	0.936 / 15.16	0.930 / 22.56
		50	0.935 / 19.14	0.928 / 20.13	0.897 / 28.28	0.931 / 19.15	0.898 / 27.02	0.885 / 38.04
		65	0.942 / 35.76	0.958 / 28.39	0.940 / 32.53	0.953 / 28.73	0.939 / 32.51	0.858 / 47.92
	AR(1)	25	0.919 / 6.23	0.940 / 7.24	0.940 / 9.35	0.942 / 10.86	0.945 / 14.80	0.945 / 21.70
		50	0.940 / 18.93	0.941 / 19.70	0.919 / 27.84	0.940 / 20.25	0.922 / 26.70	0.907 / 38.51
		65	0.945 / 34.85	0.960 / 28.73	0.943 / 31.30	0.957 / 29.05	0.947 / 31.89	0.886 / 46.37
Logistic	Frailty	25	0.896 / 13.33	0.903 / 17.05	0.889 / 22.07	0.896 / 16.71	0.899 / 21.24	0.892 / 28.47
		50	0.926 / 28.12	0.880 / 30.09	0.825 / 37.16	0.891 / 30.09	0.848 / 37.23	0.811 / 48.57
		65	0.933 / 45.16	0.946 / 38.64	0.890 / 44.48	0.940 / 38.61	0.911 / 42.29	0.793 / 58.09
	AR(1)	25	0.899 / 13.94	0.914 / 18.47	0.919 / 21.74	0.916 / 17.79	0.918 / 22.82	0.916 / 28.80
		50	0.922 / 27.75	0.903 / 29.87	0.882 / 35.30	0.914 / 28.75	0.876 / 37.86	0.855 / 48.11
		65	0.933 / 44.19	0.947 / 39.21	0.899 / 45.48	0.944 / 38.13	0.925 / 43.67	0.836 / 58.00

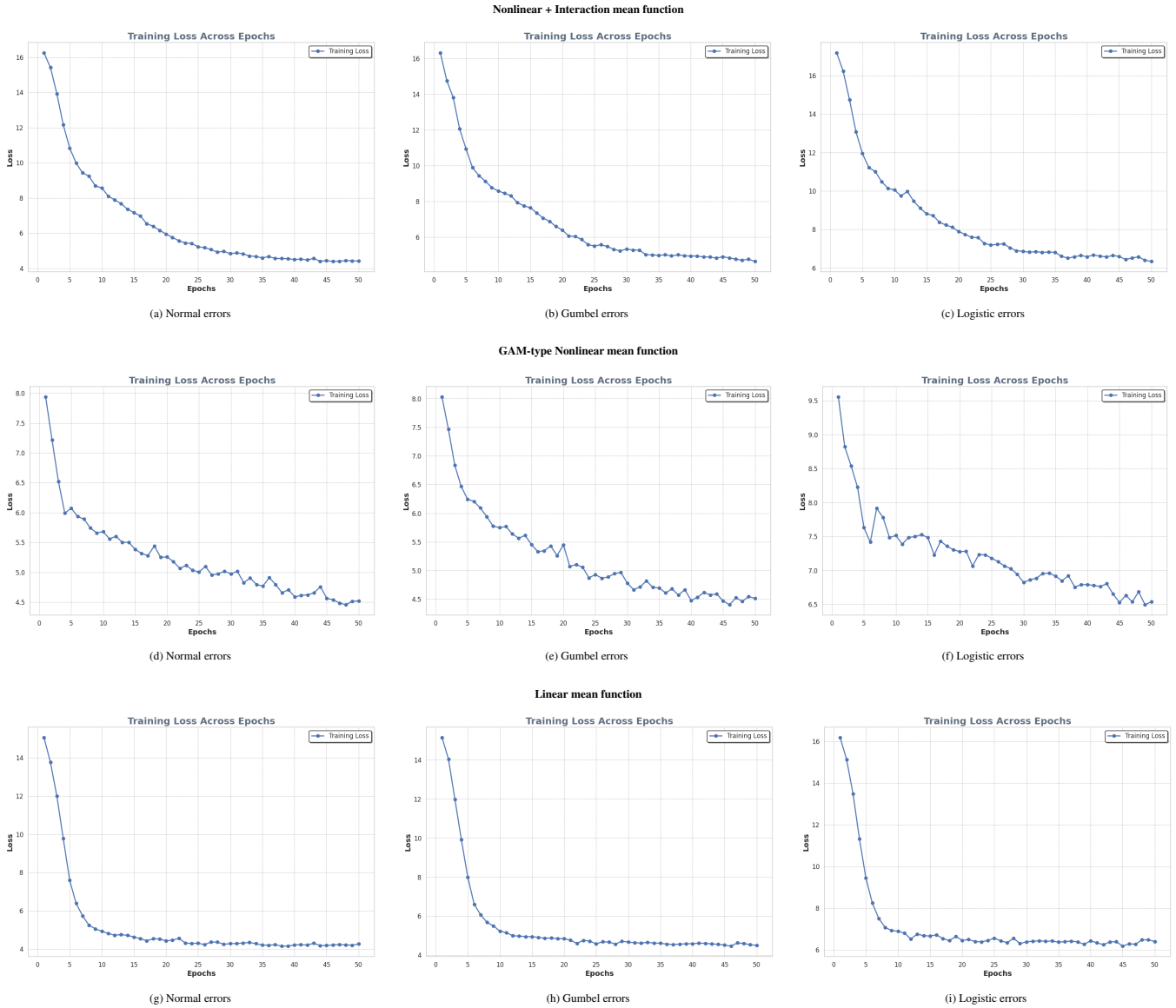
TABLE 4 | Sensitivity of RNN-AGT to sub-sampling hyperparameters. Each cell reports IPCW C-index / AMSE at the specified training epoch. Setting: nonlinear interaction mean, Normal errors, $\sigma_b = 0.5$ frailty, 25% incomplete follow-up proportion.

Hyperparameters		$n_{\text{train}} = 1{,}000$		$n_{\text{train}} = 5{,}000$	
s	b	**Epoch 5**	**Epoch 10**	**Epoch 2**	**Epoch 3**
5	32	0.851 / 9.50	0.880 / 7.43	0.880 / 6.45	0.895 / 6.65
	64	0.836 / 14.69	0.856 / 8.93	0.848 / 7.68	0.873 / 6.61
	128	0.821 / 19.08	0.840 / 14.39	0.830 / 12.69	0.843 / 8.92
10	32	0.886 / 7.57	0.917 / 7.21	0.912 / 6.10	0.934 / 6.01
	64	**0.856 / 8.77**	**0.894 / 7.38**	**0.886 / 6.51**	**0.910 / 6.02**
	128	0.840 / 14.20	0.864 / 8.43	0.857 / 7.35	0.881 / 6.56
15	32	0.901 / 7.67	0.938 / 6.48	0.934 / 6.28	0.941 / 8.80
	64	0.882 / 7.67	0.914 / 7.61	0.908 / 6.25	0.933 / 6.18
	128	0.850 / 10.24	0.886 / 7.61	0.883 / 6.58	0.904 / 6.27

Bold row: default configuration adopted in all subsequent experiments ($s = 10$, $b = 64$).

4.3.4.1 | Linear signal (Figure 3). For moderate dimensionality ($p \leq 100$), test AMSE and C-index remain competitive, with the network successfully isolating the three informative covariates from the noise background. As p grows beyond 100, test AMSE increases and the training–test concordance gap widens, reflecting the statistical challenge of recovering a three-dimensional signal from a p-dimensional feature space at fixed sample size. Crucially, the training loss trajectories (panel d) remain stable throughout, confirming that the observed performance degradation is attributable entirely to statistical generalization constraints (a deteriorating signal-to-dimension ratio) rather than any failure of the optimization procedure.

4.3.4.2 | Nonlinear signal (Figure 4). The richer nonlinear signal structure amplifies overfitting in high dimensions. The training concordance index holds steady at 0.95–0.97 regardless of p, while the test concordance declines from 0.94 at $p = 10$ to 0.67 at $p = 1{,}000$. A complementary dissociation is evident in AMSE: training AMSE decreases monotonically with p (from approximately 91.5 to 25), while test AMSE remains persistently elevated and stabilizes near 57–67 for $p \geq 300$. This hallmark pattern of flat training loss accompanied by elevated and stable test loss is diagnostic of memorization-driven overfitting, wherein the network leverages the high-dimensional parameter space to fit idiosyncratic



(a) Normal errors (b) Gumbel errors (c) Logistic errors

(d) Normal errors (e) Gumbel errors (f) Logistic errors

(g) Normal errors (h) Gumbel errors (i) Logistic errors

FIGURE 2 | Mini-batch training loss across 30 epochs under all combinations of mean function and error distribution, with $n_{\text{train}} = 1{,}000$ and 25% incomplete follow-up. Rows correspond to the interaction-based nonlinear mean function (a–c), the GAM-type nonlinear mean function (d–f), and the linear mean function (g–i). The loss decreases stably in all configurations, confirming well-conditioned optimization without divergence or oscillation across the full range of data-generating scenarios.

features of the training data rather than the underlying response surface. The non-monotone test AMSE observed for small p reflects a classical bias-variance trade-off: additional covariates carry informational content at the cost of estimation noise at fixed sample size. Taken together, RNN-AGT performs competitively for $p \leq 100$; for higher dimensions, complementary regularization strategies (dropout, weight decay, or input dimensionality reduction) are necessary to preserve generalization.

5 | Real Data Analysis

5.1 | Datasets

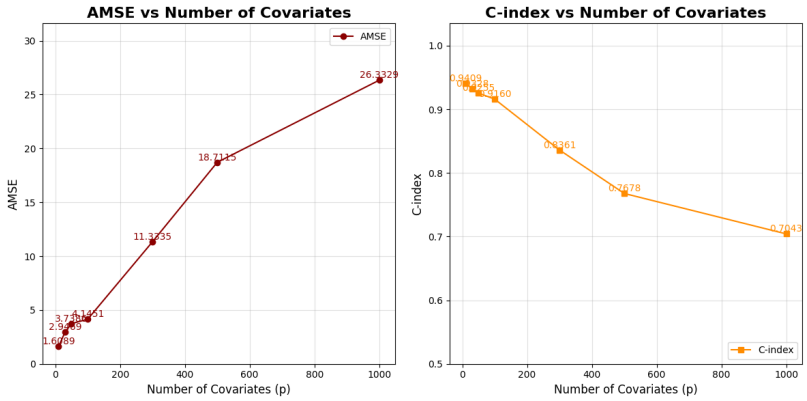
We applied RNN-AGT to two publicly available biomedical recurrent-event datasets with complementary characteristics in terms of sample size, event rate, and covariate structure.

5.1.1 | CGD Study

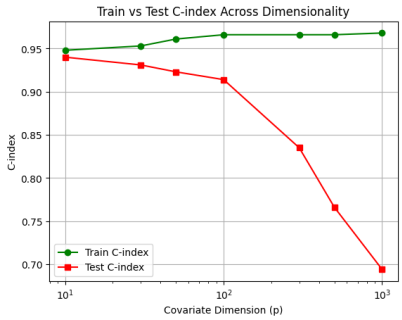
The Chronic Granulomatous Disease (CGD) dataset, originally reported in Fleming and Harrington [3] (Appendix D.2), comprises 128 patients enrolled in a randomized, placebo-controlled trial evaluating gamma interferon as prophylaxis against serious infections in immunocompromised individuals. Patients were followed from randomization to study end or administrative dropout, experiencing between 0 and 7 infection events over the course of observation, with a median follow-up of 293 days. Covariates include treatment assignment (gamma interferon versus placebo), sex, and age at enrollment.

5.1.2 | CRC Study

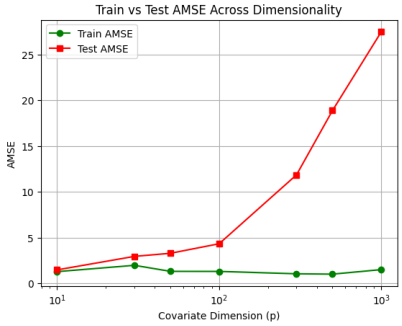
The colorectal cancer (CRC) dataset from González et al. [33], available via the `frailtypack` R package, contains 403 patients



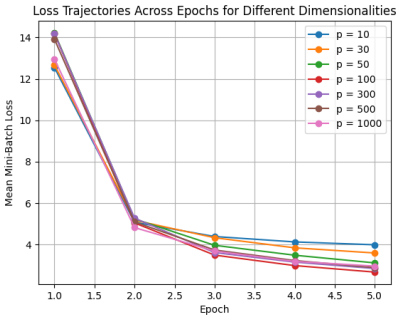
(a) AMSE (left) and IPCW C-index (right) vs. p



(b) Train vs. test C-index

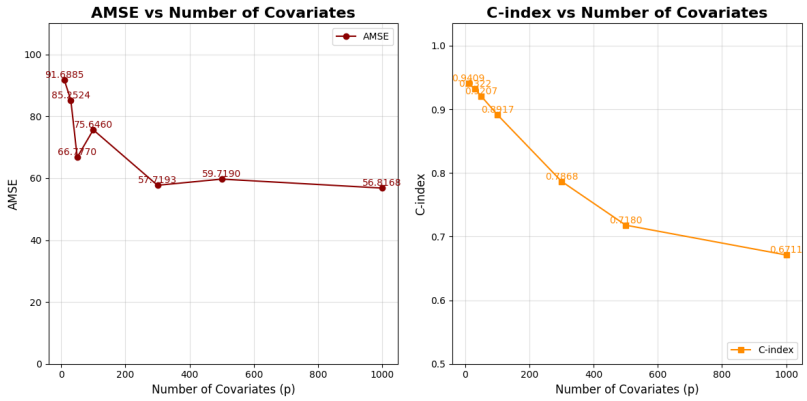


(c) Train vs. test AMSE

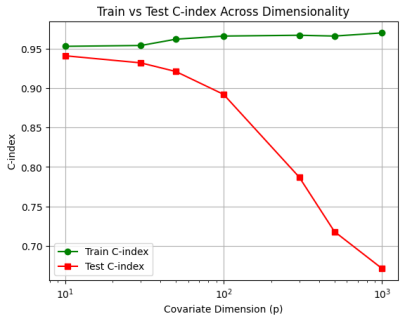


(d) Loss trajectories

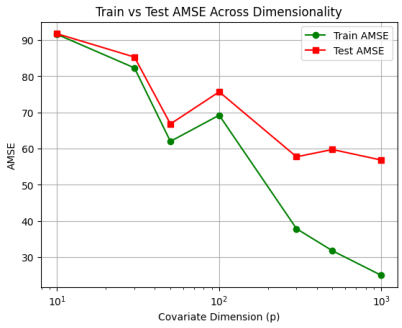
FIGURE 3 | RNN-AGT performance under the linear data-generating process as the covariate dimension p increases from 3 to 1,000. The widening train–test gap in both C-index and AMSE with growing p reflects statistical generalization constraints rather than optimization failure, as confirmed by the stability of the training loss trajectories in panel (d).



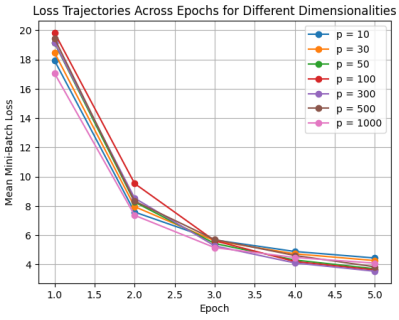
(a) AMSE (left) and IPCW C-index (right) vs. p



(b) Train vs. test C-index



(c) Train vs. test AMSE



(d) Loss trajectories

FIGURE 4 | RNN-AGT performance under the nonlinear data-generating process as p increases from 3 to 1,000. The divergence of training and test AMSE with growing p, while training concordance remains near-perfect, is the characteristic fingerprint of memorization-driven overfitting amplified by the increased structural complexity of the nonlinear signal.

monitored for hospital readmissions following colorectal cancer surgery. Follow-up ended at loss to follow-up or study termination. Three categorical or ordinal covariates were included: chemotherapy receipt (yes/no), Dukes' staging (A–B, C, or D), and sex. The larger sample size of this dataset provides a more stringent benchmark for evaluating the practical advantage of nonlinear history-dependent modeling over classical linear competitors.

5.2 | Implementation Details

For both datasets, RNN-AGT was implemented with $L = 2$ stacked GRU layers and hidden dimension $d = 64$. Pairwise sub-sampling parameters were set to $s = 30$ pairs per anchor and mini-batch size $b = 64$. Parameters were optimized using RMSprop with learning rate 3×10^{-4} over 10 training epochs on CPU. Data were partitioned $70\%/30\%$ into training and test sets using a fixed random seed to ensure reproducibility.

5.3 | Predictive Performance and Benchmarking

Table 5 summarizes the AMSE and IPCW C-index on both training and test partitions for each dataset, and Table 6 compares the test-set C-index of RNN-AGT against the gap-time and total-time PWP models and the WLW marginal model, all fitted using standard Cox regression.

On the small CGD dataset ($n = 128$), RNN-AGT achieves a test C-index of 0.631, modestly below the three Cox-based models (all at 0.648). With only 128 subjects and a $70/30$ training-test split, the effective training set contains roughly 90 observations, which is insufficient for a deep learning model to reliably exploit nonlinear structure. The observed gap of 0.017 concordance units is small in absolute terms, and both RNN-AGT and the Cox models produce concordance values only modestly above chance, suggesting that the covariates available in this dataset carry limited predictive signal regardless of the modeling approach. Differences of this magnitude, on a dataset this small, should be interpreted with caution and without strong inferential claims. The identical concordance of 0.648 across all three Cox formulations further suggests that model choice among the classical alternatives has little practical impact at this sample size. Note that the training C-index of 0.525 reported in Table 5 is only marginally above chance; as detailed in the table note, this reflects the indirect relationship between the Gehan rank loss and the concordance metric, together with the near-degenerate pair denominator arising from the small number of subjects with multiple observed events in the training partition. The test C-index of 0.631 is the operative evaluation criterion and aligns with the results of the classical competitors.

On the CRC dataset ($n = 403$), RNN-AGT achieves a test C-index of 0.637, higher than all three Cox-based competitors (PWP-GT: 0.615, PWP-TT: 0.619, WLW: 0.617). The margin of roughly 0.02 concordance units is modest, and the sample size of 403 patients, while larger, remains in the small-to-moderate range where deep learning models may not yet realize their full representational advantage. Nonetheless, the consistent direction of the difference across all three Cox competitors is encouraging and suggests that the GRU architecture is capturing some predictive structure that linear hazard models do not. These results should be viewed as a proof of concept rather than a definitive demonstration of superiority; validation on larger datasets is needed before drawing broader conclusions about the benefits of nonlinear gap-time modeling in clinical settings.

6 | Discussion

We have introduced RNN-AGT, a recurrent neural network framework that unifies the accelerated failure time paradigm with the statistical structure of recurrent gap-time data. The framework combines stacked GRU layers for sequential history-dependent encoding with a Gehan-type rank-based WRS loss that accommodates incomplete follow-up without parametric distributional assumptions, and an importance-weighted mini-batch subsampling scheme that renders training computationally tractable by reducing per-epoch complexity from $O(N^{*2})$ to $O(n's)$, a reduction of several orders of magnitude in settings with moderate to large event counts per subject. The theoretical contributions include formal proofs of the unbiasedness of the subsampled estimator (Theorem A.2) and the consistency of the IPCW concordance index as a measure of predictive discrimination in the recurrent gap-time setting (Theorem A.3), together with the unified Gehan-type WRS loss for nonlinear predictors (Theorem A.1).

The simulation results reveal several patterns of broad methodological interest. Most strikingly, the training loss decreases stably and without oscillation across all configurations spanning three mean functions, three error distributions, two incomplete follow-up regimes, and two within-subject dependence structures, confirming that the mini-batch Gehan objective provides well-conditioned gradients for stochastic optimization despite the non-convexity of the nonlinear loss landscape. The concordance index and AMSE respond characteristically differently to extended training: the C-index typically stabilizes after a moderate number of epochs, while AMSE continues to inflate, reflecting the tendency of rank-based objectives to preserve relative ordering while the squared-error component overfits to the magnitude scale. This decoupling has an important practical implication: models selected to maximize concordance on a validation set, rather than to minimize squared prediction error, may be preferred in discrimination-oriented applications, and monitoring both metrics jointly during training is strongly recommended.

The high-dimensional experiments reveal an informative asymmetry between linear and nonlinear signal structures. In the linear case, growing p inflates estimation variance without introducing additional signal, degrading both training and test performance in tandem. In the nonlinear case, RNN-AGT exploits the high-dimensional parameter space aggressively to reduce training AMSE, inducing overfitting: a widening train–test gap that constitutes the canonical signature of the bias-variance trade-off under high dimensionality. Crucially, in both settings optimization itself remains stable and well-behaved regardless of p, establishing that performance degradation in high dimensions reflects statistical generalization constraints rather than any structural limitation of the GRU architecture or the Gehan-type objective.

The present work has several limitations that motivate future research directions. First, the current implementation does not explicitly handle time-varying covariates, though the architecture accommodates them naturally by redefining the sequential input as

TABLE 5 | Predictive performance of RNN-AGT on the CGD and CRC datasets, evaluated by IPCW C-index and AMSE on training and test partitions.

Dataset	Training set		Test set	
	C-index	AMSE	C-index	AMSE
CGD Study ($n = 128$)[a]	0.525	41.24	0.631	28.31
CRC Study ($n = 403$)	0.618	63.31	0.637	42.95

[a]The low training C-index for the CGD study reflects two compounding factors. First, the Gehan-type rank loss optimises the relative ordering of residuals across the *full* training set rather than maximising the concordance index directly; these objectives need not be monotonically related during early training. Second, the effective training set of approximately 90 subjects contains very few subjects with more than one observed infection, producing a near-degenerate concordance denominator (few comparable pairs) on which Monte Carlo variability dominates. The substantially higher test C-index of 0.631 is the operative evaluation metric and is comparable across methods (Table 6).

TABLE 6 | Test-set IPCW C-index comparison of RNN-AGT and three established Cox-based recurrent-event models on both datasets. The best value in each column is shown in **bold**.

Model	CGD ($n = 128$)	CRC ($n = 403$)
WLW	**0.648**	0.617
PWP-TT	**0.648**	0.619
PWP-GT	**0.648**	0.615
RNN-AGT	0.631	**0.637**

WLW = Wei–Lin–Weissfeld; PWP-TT = Prentice–Williams–Peterson total-time;

PWP-GT = Prentice–Williams–Peterson gap-time.

described in Section 3.2; a thorough empirical investigation of this extension is left to future work. Second, large-sample theory for the nonlinear estimator $\hat{\mathbf{w}}$ that minimizes $L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w})$ (including consistency, rates of convergence, and distributional approximations for uncertainty quantification) remains an open theoretical problem whose resolution would strengthen the inferential foundations of RNN-AGT. Third, in truly high-dimensional settings ($p \gg 100$), complementary strategies are essential: variable selection via group lasso or elastic-net penalties on the input weights, low-dimensional embedding through autoencoders or principal component analysis prior to GRU encoding, or attention-based feature selection mechanisms.

Promising architectural extensions include replacing the GRU layers with transformer encoders [26], which exploit multi-head self-attention to capture non-sequential long-range dependencies that may be difficult to propagate through recurrent hidden states. Hybrid loss functions combining rank-based objectives with regression-based calibration penalties offer a principled path to jointly optimizing discrimination and absolute prediction accuracy. Finally, adaptive sub-sampling strategies based on importance weights proportional to the magnitude of pairwise loss contributions [34] may further accelerate convergence and reduce variance in gradient estimates, enabling computationally efficient RNN-AGT training even at biobank scales.

FINANCIAL DISCLOSURE

None reported.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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APPENDIX

A | Technical Proofs

We collect the three main theoretical results supporting RNN-AGT. Throughout, $\mathcal{O} = \{\mathcal{O}_i\}_{i=1}^n$ denotes the full observed dataset.

Theorem A.1 (Nonlinear Gehan-type WRS loss)**.** Let $\widehat{\mu}_{ij}(\mathbf{w})$ denote the history-dependent nonlinear predictor produced by RNN-AGT with parameters $\mathbf{w} \in \mathcal{W}$, and define the induced residual $e_{ij}(\mathbf{w}) = Y_{ij} - \widehat{\mu}_{ij}(\mathbf{w})$. The Gehan-type WRS training loss for the nonlinear AFT gap-time model is

$$L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \sum_{l=1}^{n} \sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} \big[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w})\big]^{-}. \tag{A1}$$

When the predictor is restricted to the linear form $\widehat{\mu}_{ij}(\mathbf{w}) = \theta' \mathbf{Z}_i$, equation (A1) reduces to the WRS Gehan loss (3.4). When $K_i^* \equiv 1$, it reduces to the single-event Gehan-type rank loss of Kim, Park, and Kang [22].

Proof. The Gehan-type WRS objective (3.4) depends on θ only through the residuals $\{e_{ij}(\theta)\}$. Replacing the linear residuals $e_{ij}(\theta) = Y_{ij} - \theta' \mathbf{Z}_i$ by the induced nonlinear residuals $e_{ij}(\mathbf{w}) = Y_{ij} - \widehat{\mu}_{ij}(\mathbf{w})$ throughout (3.4) yields (A1) directly. The two reduction claims follow immediately by setting $\widehat{\mu}_{ij}(\mathbf{w}) = \theta' \mathbf{Z}_i$ and $K_i^* \equiv 1$, respectively. □

Remark 3. While $L_{\mathrm{G}}(\theta)$ is convex in θ by the positive-part/negative-part structure of the Gehan pairwise terms, $L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w})$ is generally non-convex in $\mathbf{w}$ because the residual map $\mathbf{w} \longmapsto e_{ij}(\mathbf{w})$ is a nonlinear function of the network parameters. This non-convexity necessitates stochastic gradient methods with appropriate regularization rather than convex solvers or linear programming, a practical trade-off accepted in exchange for the considerably greater representational capacity of the nonlinear predictor.

Theorem A.2 (Unbiasedness of the sub-sampled loss)**.** Fix $s \geq 1$. For each $(i, j) \in \mathcal{U}$, let $(l, k)_{ij,1}, \ldots, (l, k)_{ij,s}$ be i.i.d. draws from a sampling distribution q_{ij} on $\mathcal{I}$ satisfying $q_{ij}(l, k) > 0$ for all $(l, k) \in \mathcal{I}$. Define the sub-sampled estimator $\widetilde{L}_{\mathrm{G},s}^{\mathrm{NL}}(\mathbf{w})$ as in (3.5). Then, for every $\mathbf{w} \in \mathcal{W}$,

$$\mathbb{E}\big[\widetilde{L}_{\mathrm{G},s}^{\mathrm{NL}}(\mathbf{w}) \mid \mathcal{O}\big] = L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w}).$$

In particular, when q_{ij} is the uniform distribution on $\mathcal{I}$, the importance weights are constant and $\widetilde{L}_{\mathrm{G},s}^{\mathrm{NL}}(\mathbf{w})$ simplifies to a scaled average of $m = n's$ sampled pair contributions.

Proof. Write the full loss as

$$L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w}) = \frac{1}{n} \sum_{(i,j)\in\mathcal{U}} \sum_{(l,k)\in\mathcal{I}} a_{ij}(l,k;\mathbf{w}),$$

where $a_{ij}(l,k;\mathbf{w}) = K_i^{*-1} K_l^{*-1} \big[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w})\big]^-$. For fixed $(i,j) \in \mathcal{U}$, draw $(l,k)_{ij,r}$ i.i.d. from q_{ij} and define the importance-weighted single-draw contribution $X_{ij,r}(\mathbf{w}) = a_{ij}((l,k)_{ij,r};\mathbf{w})/q_{ij}((l,k)_{ij,r})$. Since $q_{ij}(l,k) > 0$ for all $(l,k) \in \mathcal{I}$,

$$\mathbb{E}[X_{ij,r}(\mathbf{w}) \mid \mathcal{O}] = \sum_{(l,k)\in\mathcal{I}} q_{ij}(l,k) \frac{a_{ij}(l,k;\mathbf{w})}{q_{ij}(l,k)} = \sum_{(l,k)\in\mathcal{I}} a_{ij}(l,k;\mathbf{w}).$$

By the i.i.d. sampling structure and linearity of conditional expectation,

$$\begin{aligned}\mathbb{E}\big[\widetilde{L}_{\mathrm{G},s}^{\mathrm{NL}}(\mathbf{w}) \mid \mathcal{O}\big] &= \frac{1}{n} \sum_{(i,j)\in\mathcal{U}} \mathbb{E}\left[\frac{1}{s}\sum_{r=1}^{s} X_{ij,r}(\mathbf{w}) \,\middle|\, \mathcal{O}\right] \\ &= \frac{1}{n} \sum_{(i,j)\in\mathcal{U}} \sum_{(l,k)\in\mathcal{I}} a_{ij}(l,k;\mathbf{w}) = L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w}).\end{aligned}$$

Under $q_{ij}(l,k) = 1/N^*$, the importance weight $1/q_{ij}$ is the constant N^*, so (3.5) reduces to $N^*/(n\,s)\sum_r [e_{(l,k)_r}(\mathbf{w}) - e_{ij}(\mathbf{w})]^-$, an unweighted average of m sampled contributions scaled by a global constant. □

Theorem A.3 (Consistency of the IPCW concordance index)**.** Suppose that $C_i \perp\!\!\!\perp \{T_{ij}, j \geq 1\} \mid \mathbf{Z}_i$ (non-informative end-of-monitoring: the monitoring time C_i is independent of the gap-time process given covariates) and that $\widehat{KM}(t) \xrightarrow{p} G_C(t)$ uniformly on the support of the event-time comparisons. Then the IPCW concordance index $\widehat{C}_{\mathrm{IPCW}}$ defined in (4.1) satisfies

$$\widehat{C}_{\mathrm{IPCW}} \xrightarrow{p} \mathbb{P}\big(\widehat{\mu}_{ij} < \widehat{\mu}_{kl} \mid T_{ij} < T_{kl}\big),$$

so $\widehat{C}_{\mathrm{IPCW}}$ is a consistent estimator of the true concordance probability among cross-subject gap-time pairs and therefore constitutes a valid measure of predictive discrimination under incomplete monitoring.

Proof. By Bayes' theorem,

$$\mathbb{P}\big(\widehat{\mu}_{ij} < \widehat{\mu}_{kl} \mid T_{ij} < T_{kl}\big) = \frac{\mathbb{E}\big[\mathbf{1}\{T_{ij} < T_{kl}\}\mathbf{1}\{\widehat{\mu}_{ij} < \widehat{\mu}_{kl}\}\big]}{\mathbb{E}\big[\mathbf{1}\{T_{ij} < T_{kl}\}\big]}.$$

Under non-informative end-of-monitoring, the probability that the jth gap time of subject i is fully observed satisfies $\mathbb{P}(\delta_{ij} = 1 \mid S_{ij}) = G_C(S_{ij})$, hence $\mathbb{E}[\delta_{ij}/G_C(S_{ij}) \mid S_{ij}] = 1$. An application of the law of total expectation then shows that replacing $\mathbf{1}\{T_{ij} < T_{kl}\}$ by its IPCW-reweighted surrogate $\delta_{ij}\mathbf{1}\{G_{ij} < G_{kl}\}/G_C(S_{ij})$ preserves the expectation in both the numerator and denominator of $\widehat{C}_{\mathrm{IPCW}}$. Substituting the Kaplan–Meier estimator $\widehat{KM}$ for the unknown G_C and invoking the law of large numbers for U-statistics (see van der Vaart 35, Chapter 12) in the numerator and denominator separately, together with Slutsky's theorem applied to their ratio, yields the stated convergence in probability. □